

Aircraft Performance for Propulsion System Engineers

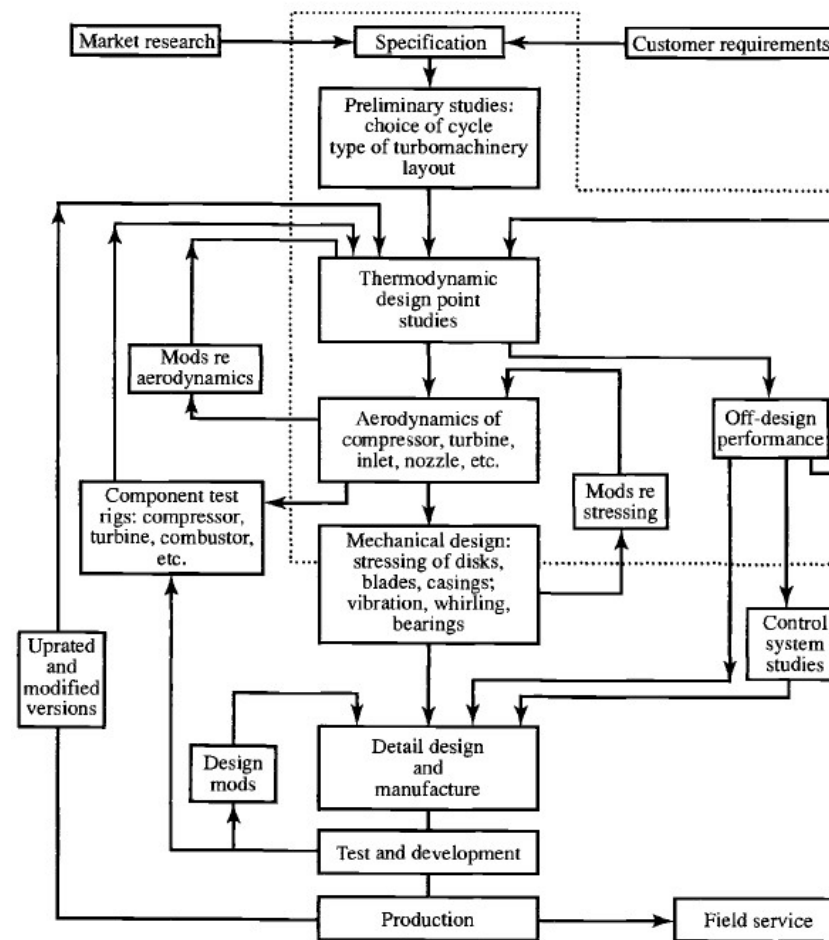
Class 3

Objective

- The objective of this class is to understand the different propulsion systems and their effect on the forces of the aircraft as well as the physics behind the generation of those forces.
- Understand the concept of Thrust and different parameters that affect the aircraft and engine performance.

Aircraft Engine Design

- There is no absolute roadmap for the design of a gas turbine engine. The design procedure for a typical gas turbine engine, shown in the picture, requires that thermodynamic design point studies and off design performance be the initial steps of design.

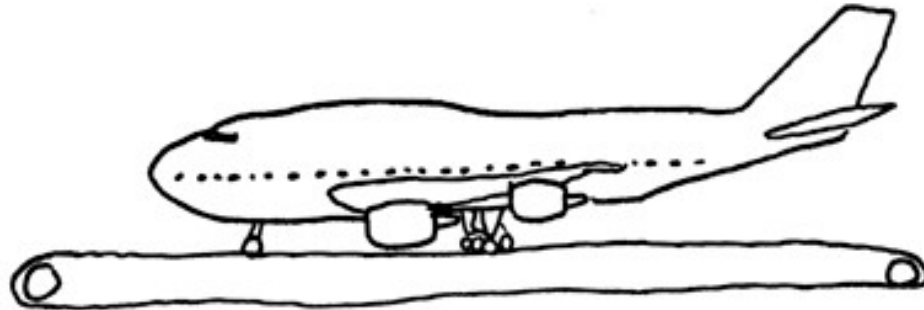


Engine Thrust

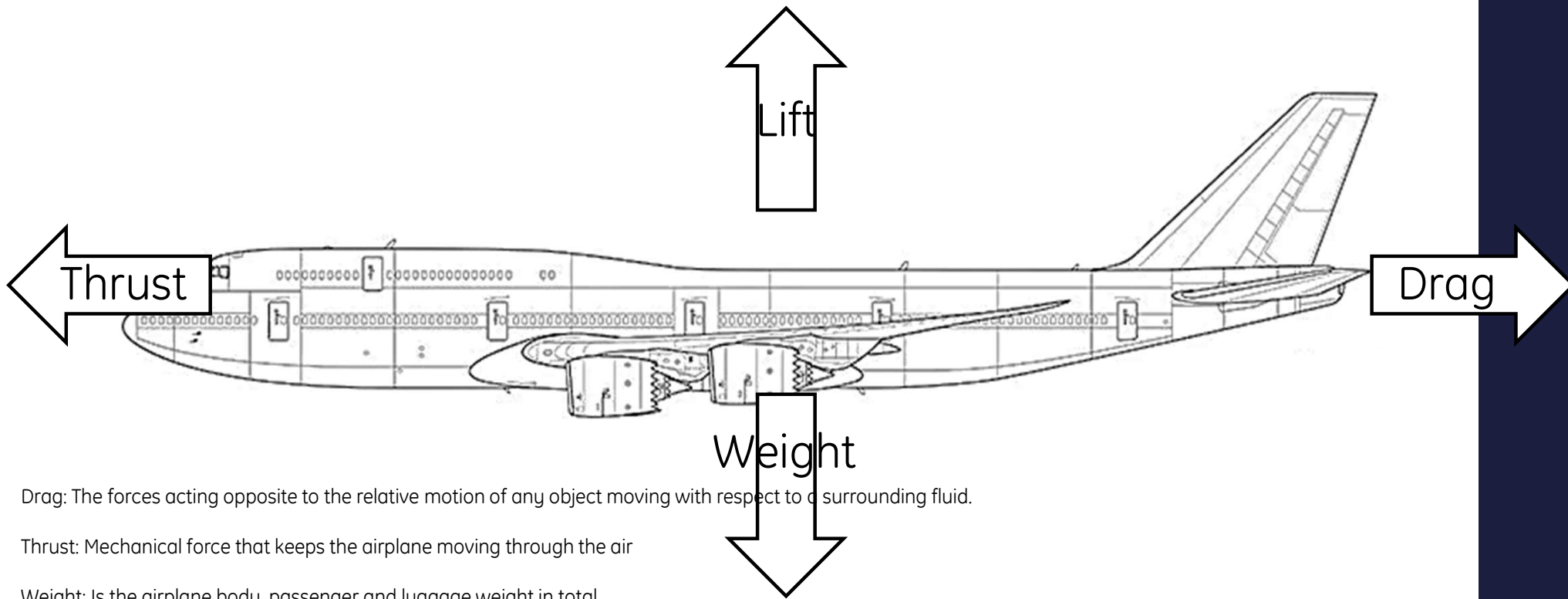
Physics

Aerodynamics of an Airplane

- Imagine a 747 is sitting on a conveyor belt, as wide and long as a runway. The conveyor belt is designed to exactly match the speed of the wheels, moving in the opposite direction.
- Can the plane take off?



Aerodynamics of an Airplane



Drag: The forces acting opposite to the relative motion of any object moving with respect to a surrounding fluid.

Thrust: Mechanical force that keeps the airplane moving through the air

Weight: Is the airplane body, passenger and luggage weight in total

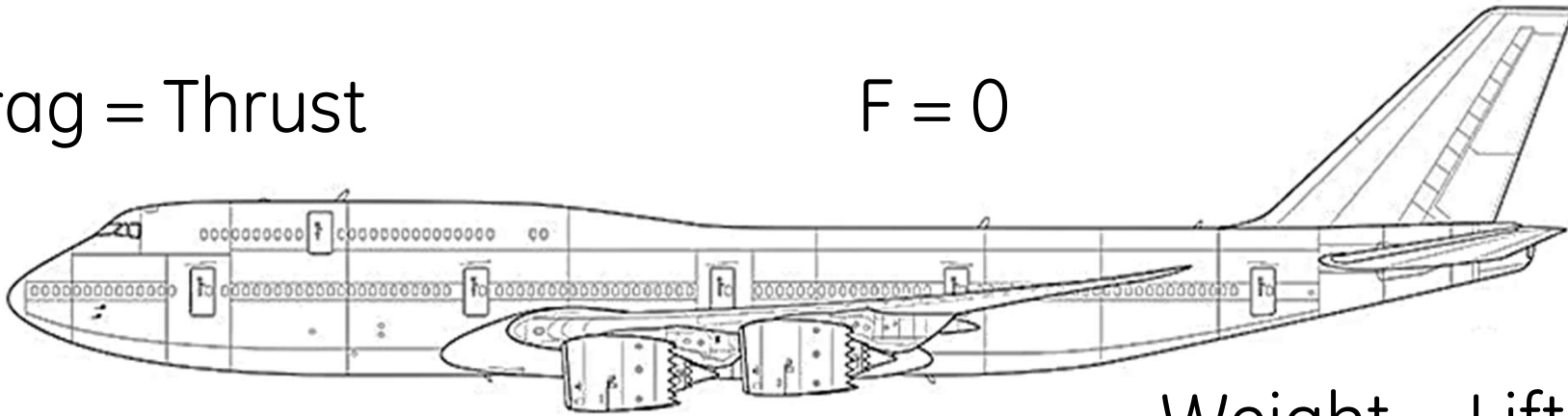
Lift: Overcomes the weight and holds up the airplane in the air. It is created mostly by wings to keep the planes aloft.

Aerodynamics of an Airplane

So to keep the airplane moving, flying straight and leveled what conditions do we need?

Drag = Thrust

$$F = 0$$



Weight = Lift

- Drag > Thrust = Plane Slows Down
- Thrust > Drag = Plane Moves Faster
- Weight > Lift = Plane Descends
- Lift > Weight = Plane Climbs

Engine Thrust

- To derive an expression for the engine thrust, it is most convenient to describe a control volume surrounding the engine and apply momentum principles to the fluid flow crossing the boundaries of the control volume.

$$\sum (\dot{m}V_x)_{out} - \sum (\dot{m}V_x)_{in} = \sum (F_x)_{fluid}$$

$$\sum (\dot{m}V_x)_o = (\dot{m}_0 + \dot{m}_f)V_9 + [\rho_0 V_0 (A - A_9)]V_0$$

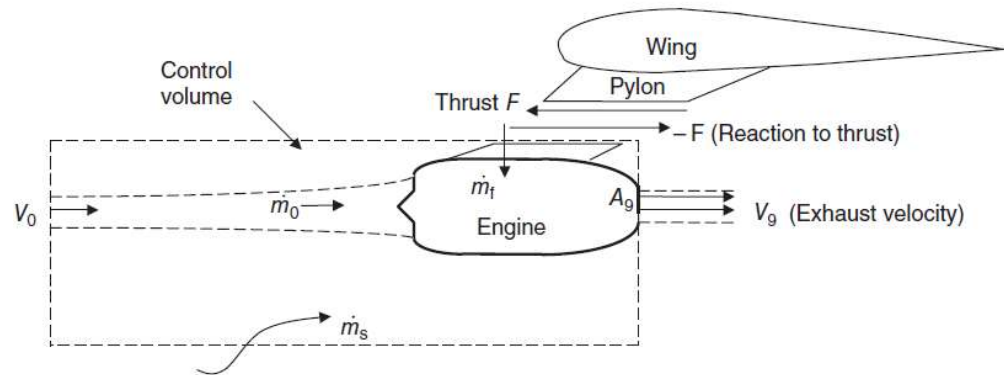
$$\sum (\dot{m}V_x)_{in} = (\rho_0 V_0 A)V_0 + \dot{m}_s V_0$$

$$\sum (F_x)_{fluid} = (-F)_{fluid} - (p_9 - p_0)A_9$$

$$(\dot{m}_0 + \dot{m}_f)V_9 - \dot{m}V_0 = (-F_x)_{fluid} - (p_9 - p_0)A_9$$

$$F_n = (\dot{m}_0 + \dot{m}_f)V_9 + (p_9 - p_0)A_9 - \dot{m}V_0$$

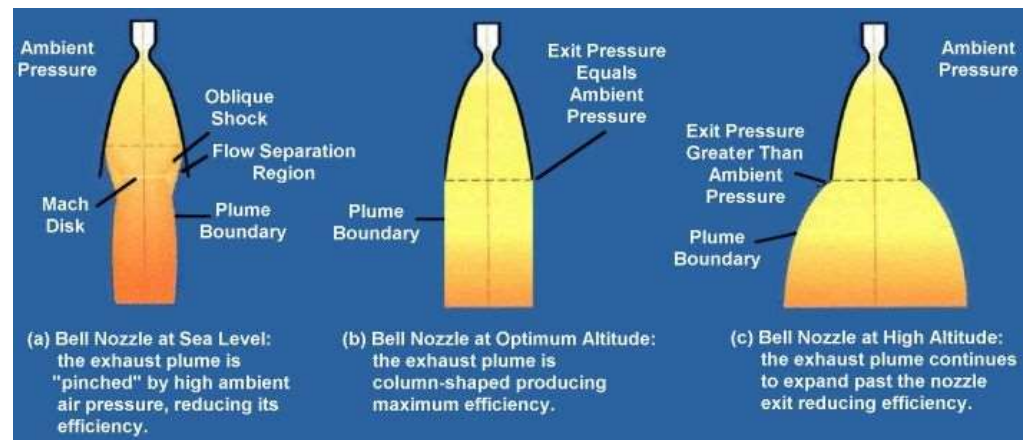
What does each term of the equations mean?



Expansion of the flow

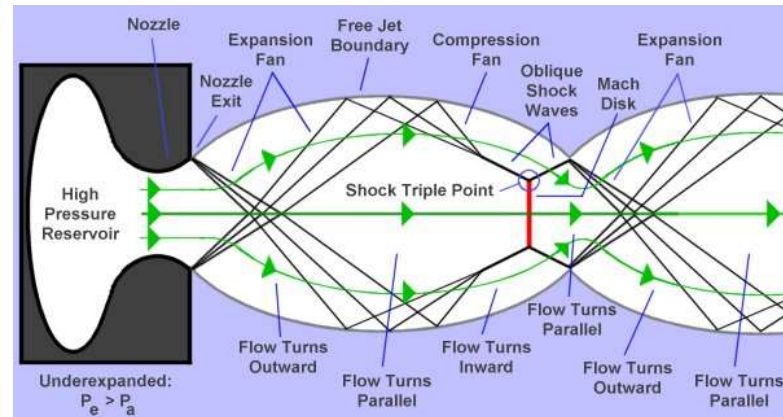
- Expansion is the process that converts the thermal energy of the hot gases into kinetic energy to move an object forward. In other words, the hot gases created by burning fuel inside a jet or rocket engine are exhausted through a nozzle to produce thrust.
- It is the shape of this nozzle that is key to the expansion process. As that high temperature flow is exhausted, it expands against the walls of the nozzle to create a force that pushes the vehicle forward.

- If $p_9 < p_0$ = overexpanded nozzle
- If $p_9 = p_0$ = perfectly expanded nozzle
- If $p_9 > p_0$ = underexpanded nozzle



100

Underexpanded

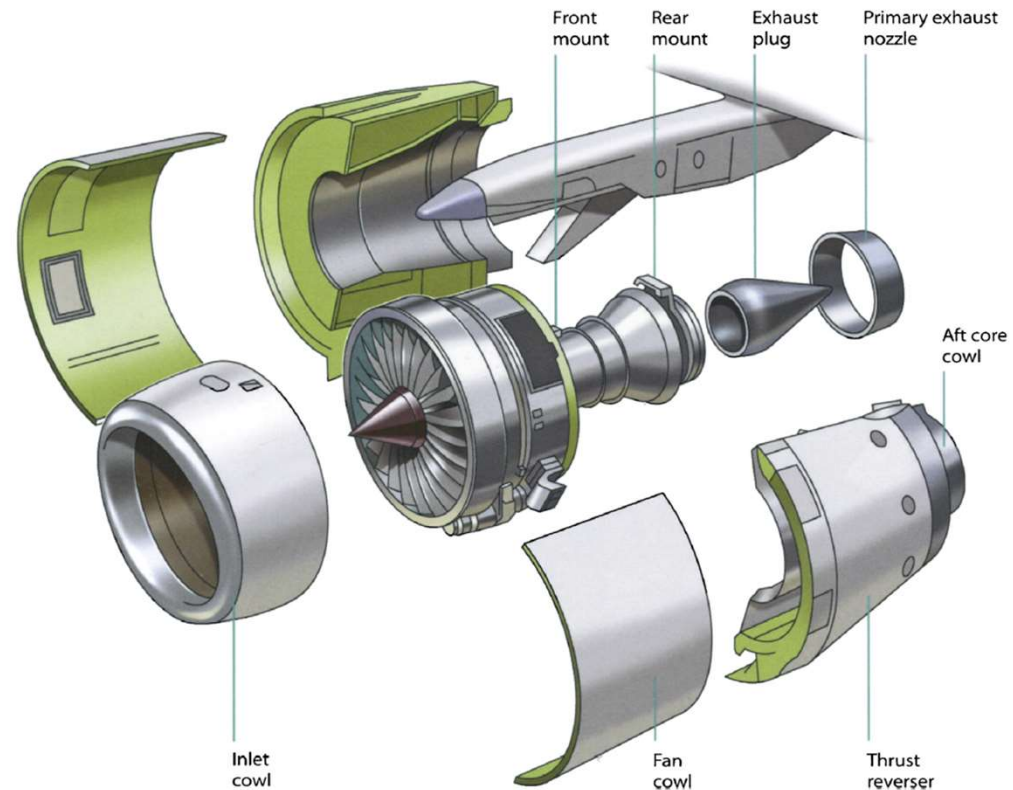


Installed Thrust vs Uninstalled Thrust

A nacelle is a streamlined enclosure that fits around a dressed engine and interfaces with the aircraft structure.

The primary objectives of a nacelle are to

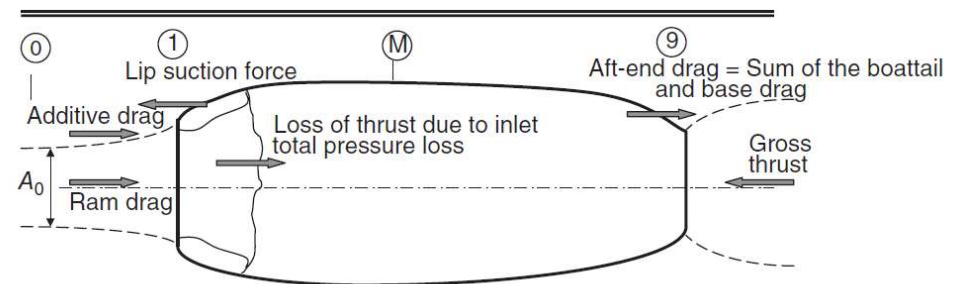
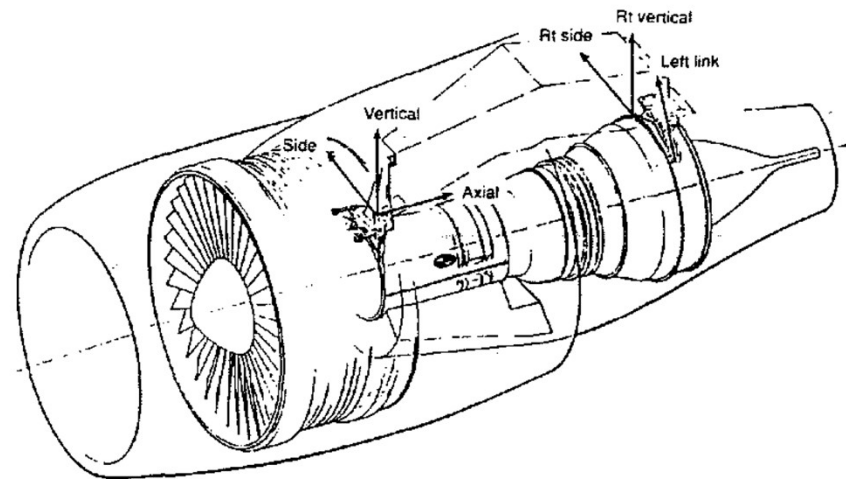
- Provide low drag
- Ensure good engine performance throughout the aircraft flight and ground envelopes
- Reduce engine noise with acoustic treatment of nacelle structure
- Prevent ice impact damaging fan blades



Installed vs Uninstalled Thrust

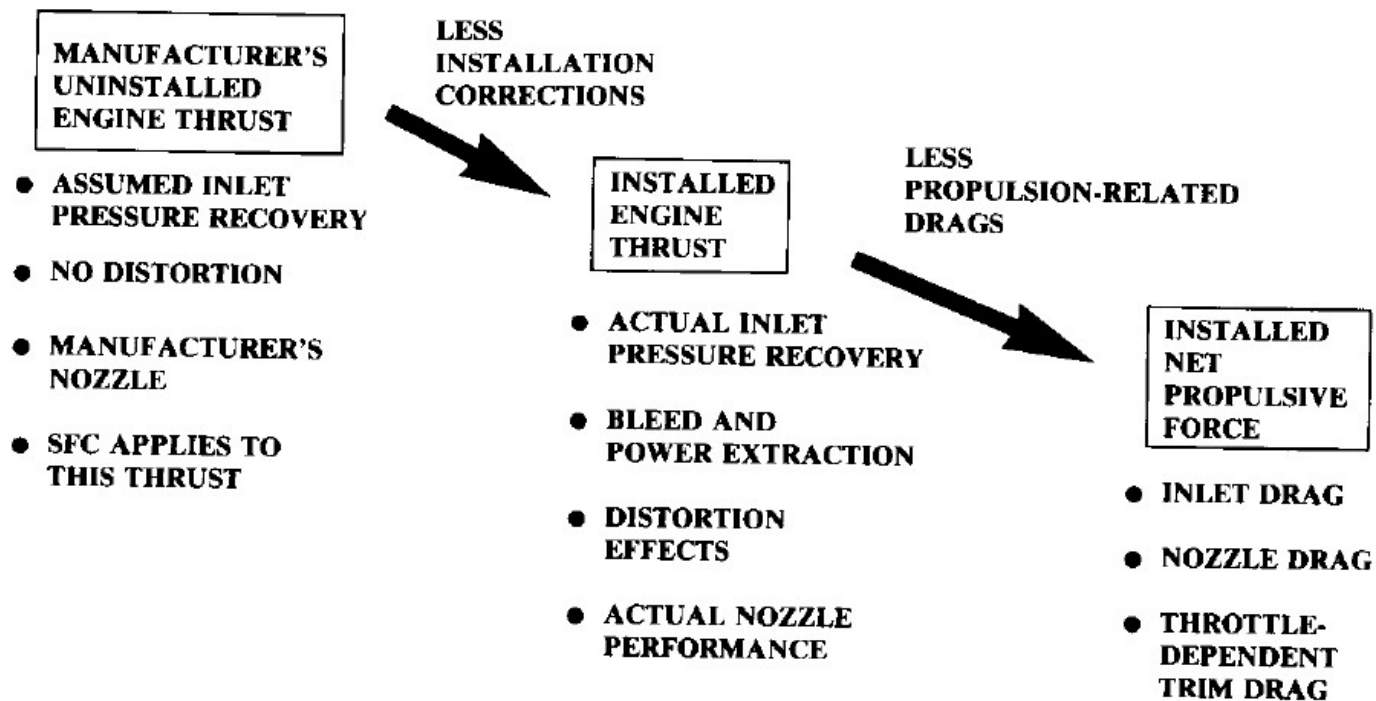
- As indicated by the description of the terms “installed thrust” and “uninstalled thrust”, they refer to the actual propulsive force transmitted to the aircraft by the engine and the thrust produced by the engine if it had zero external losses respectively.

$$F_{installed} = F_{uninstalled} - D_{nacelle}$$



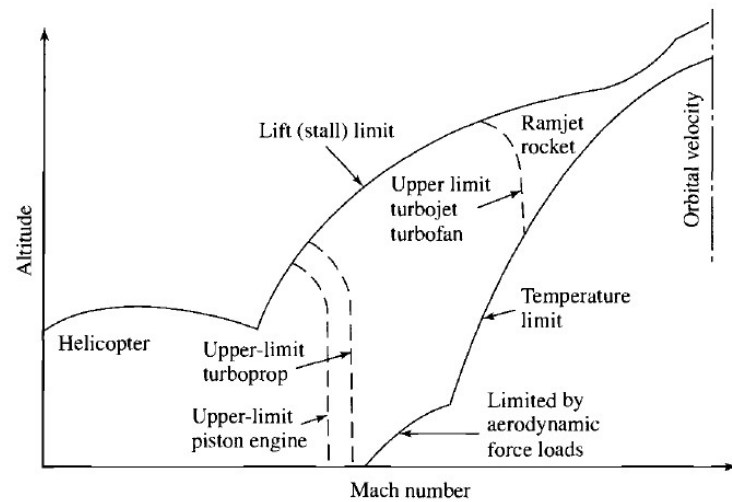
Net propulsive thrust	=	Gross thrust	-	Ram drag	-	Drag due to inlet total pressure loss	-	Additive drag	+	Lip suction force	-	Aft-end drag
$F_{n,inst.} = \underbrace{F_{gross} - D_{ram}}_{\text{Uninstalled net thrust}} - \Delta F_{inlet, pt} - \underbrace{D_{add} + F_{lip}}_{\text{Spillage drag, } D_{spillage}} - D_{aft-end}$												

Installed vs Uninstalled Thrust

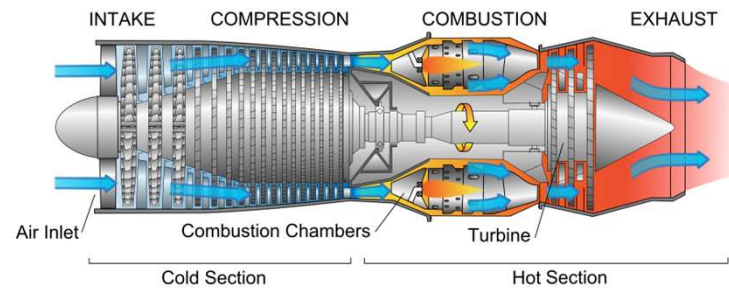


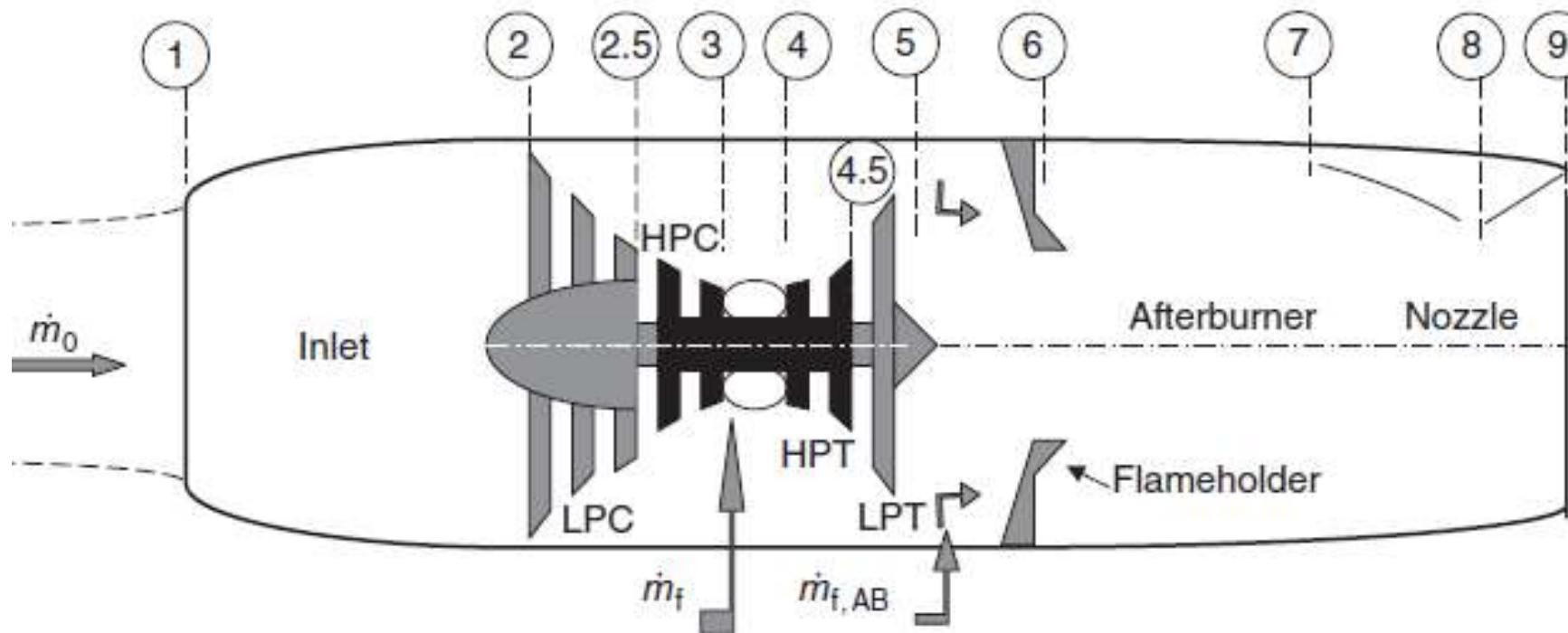
Thrust constraints – Technology selection

Operational Limits



Internal Physics





Engine Station Nomenclature

Engine Performance Parameters

Engine Performance Parameters

Specific Thrust

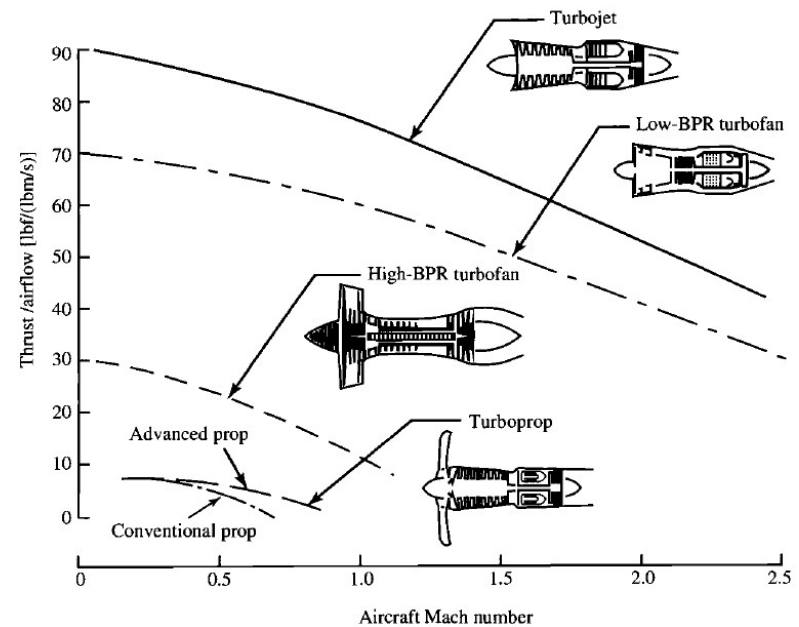
- The size of the air intake system is a design parameter that establishes the flow rate of air. Accordingly, the fuel pump is responsible for setting the fuel flow rate in the engine. In general, the magnitude of the thrust produced is directly proportional to the mass flow rates of the fluid flow through the engine.

Airbreathing engine performance parameter # 1:

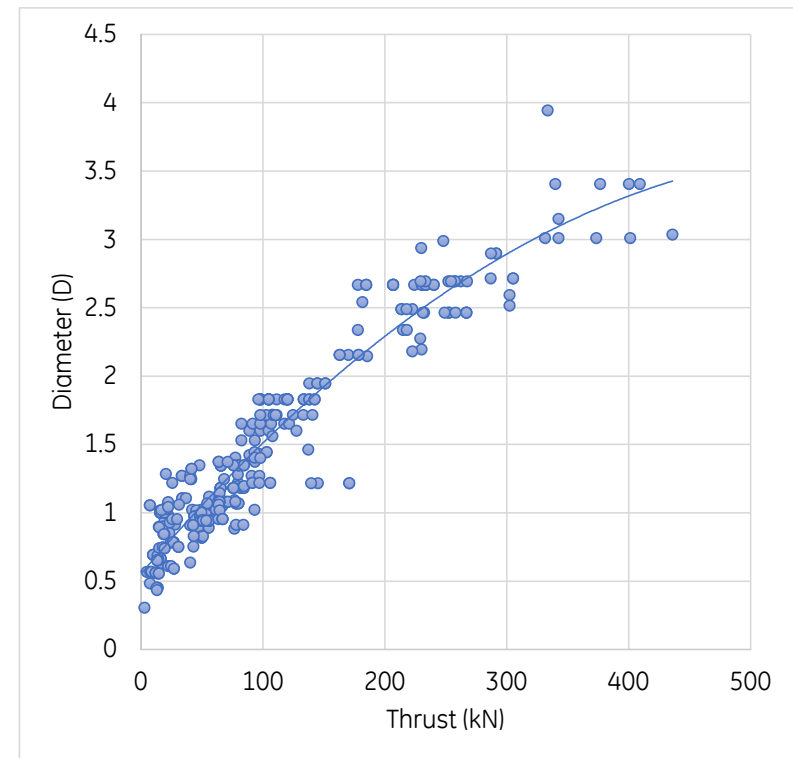
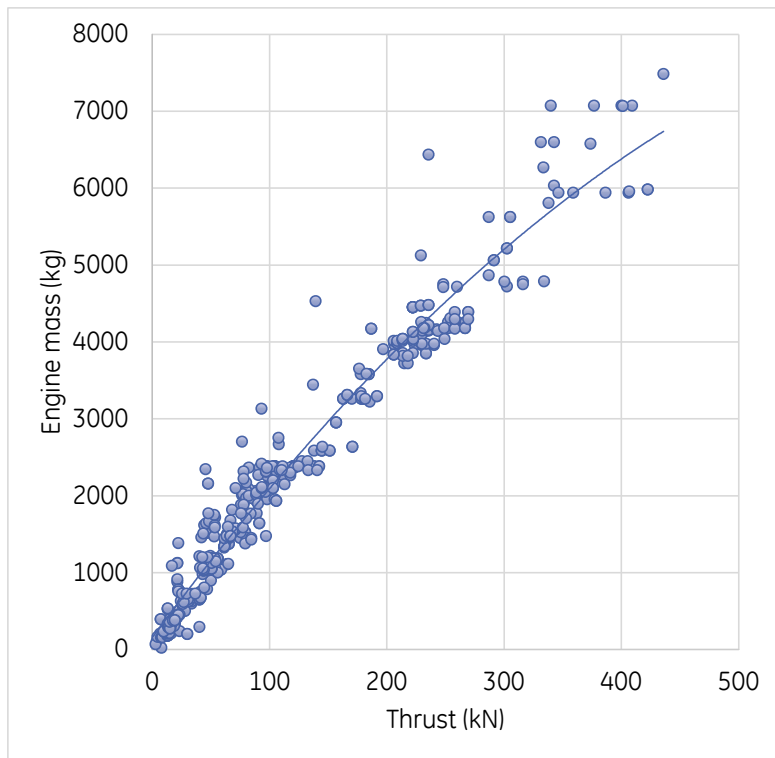
$\frac{F}{\dot{m}_0}$ “specific thrust” with a metric unit of [N · s/kg] and a British unit of [lbf · s/lbm]

Engine Performance Parameters

- High specific thrust reduces the cross-sectional area of the engine and has a direct influence on engine weight and installation losses.



Engine Performance Parameters



Engine Performance Parameters

Specific Fuel Consumption

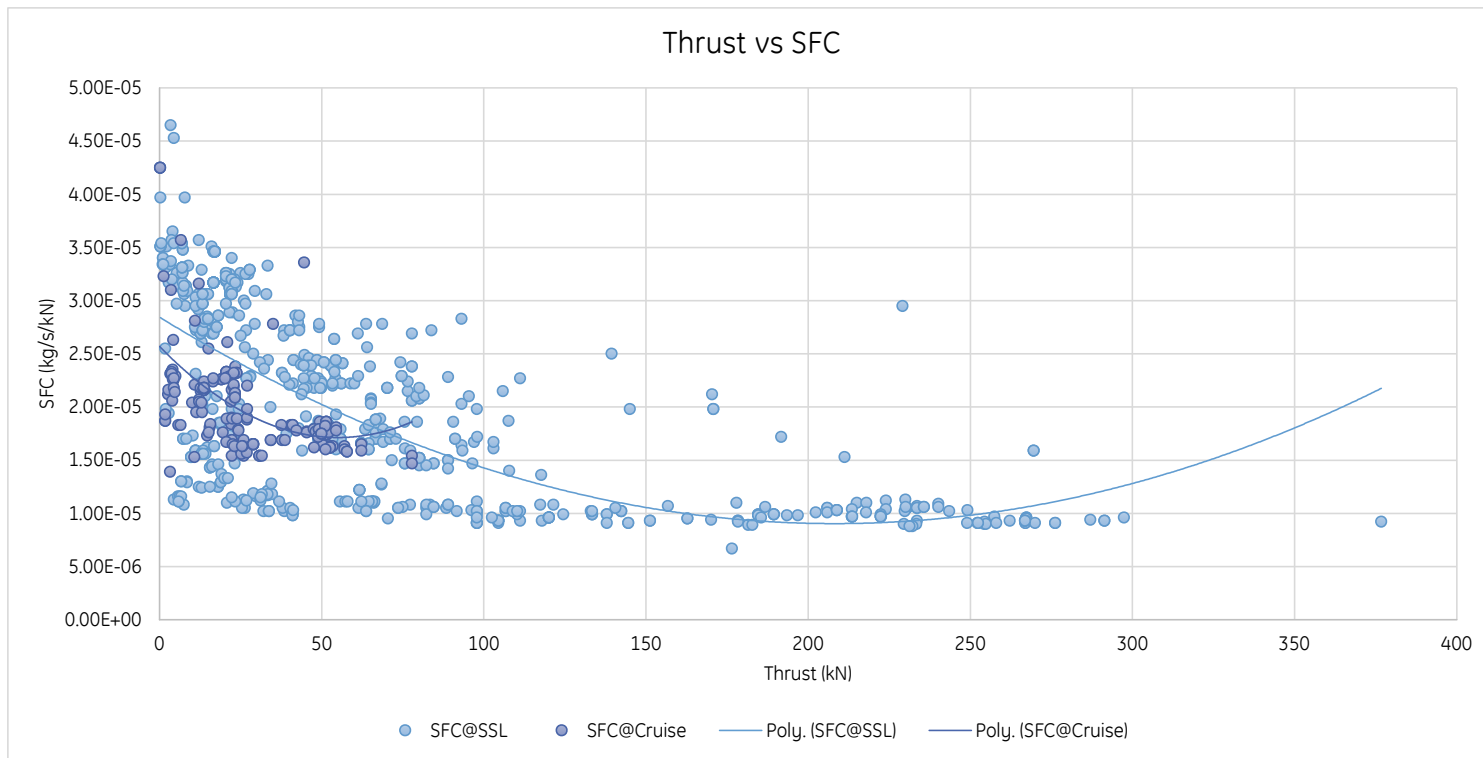
- The ability to produce thrust with a minimum of fuel expenditure is another parameter that is considered to be a performance parameter in an engine. For an airbreathing engine, the ratio of fuel flow rate per unit thrust force produced is called thrust-specific fuel consumption (TSFC), or sometimes just the specific fuel consumption (sfc) and is defined as

Airbreathing engine performance parameter # 2:

$$\frac{\dot{m}_f}{F} \text{ “thrust-specific fuel consumption”}$$

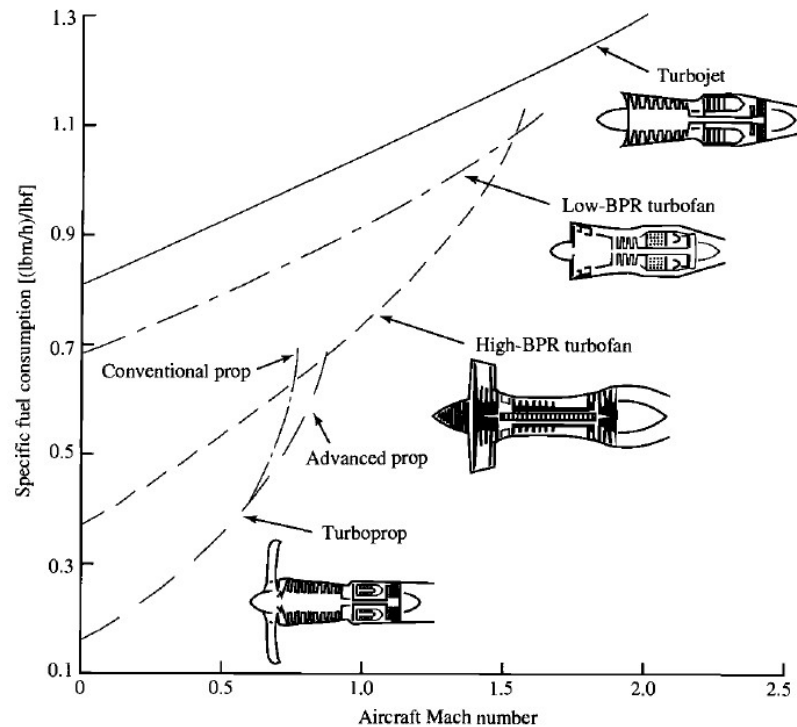
(TSFC) with a metric unit of [mg/s/N] and a British unit of : [lbm/h/lbf]

Engine Performance Parameters



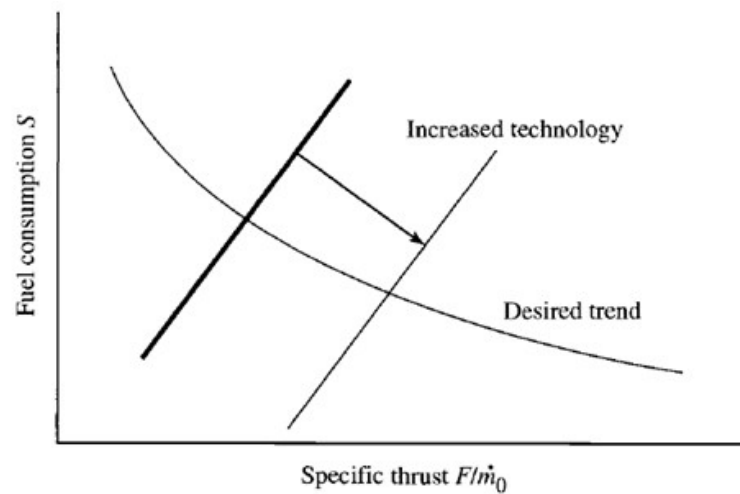
Engine Performance Parameters

Low engine fuel consumption can be directly translated into longer range, increased payload and/or reduced aircraft size.



Engine Performance Parameters

- Aircraft manufacturers desire engines having low thrust specific fuel consumption and high specific thrust.

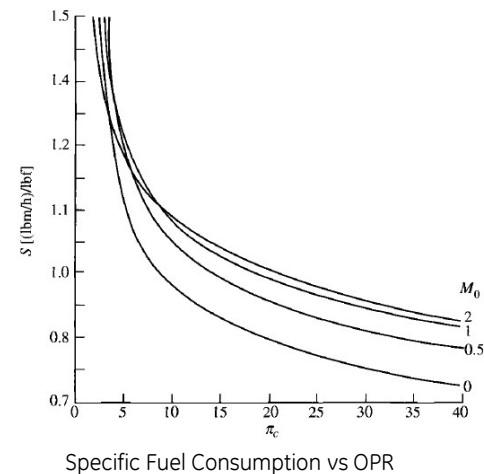
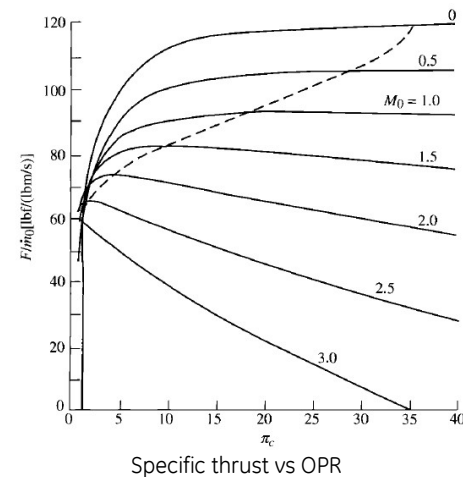


Engine Performance Parameters

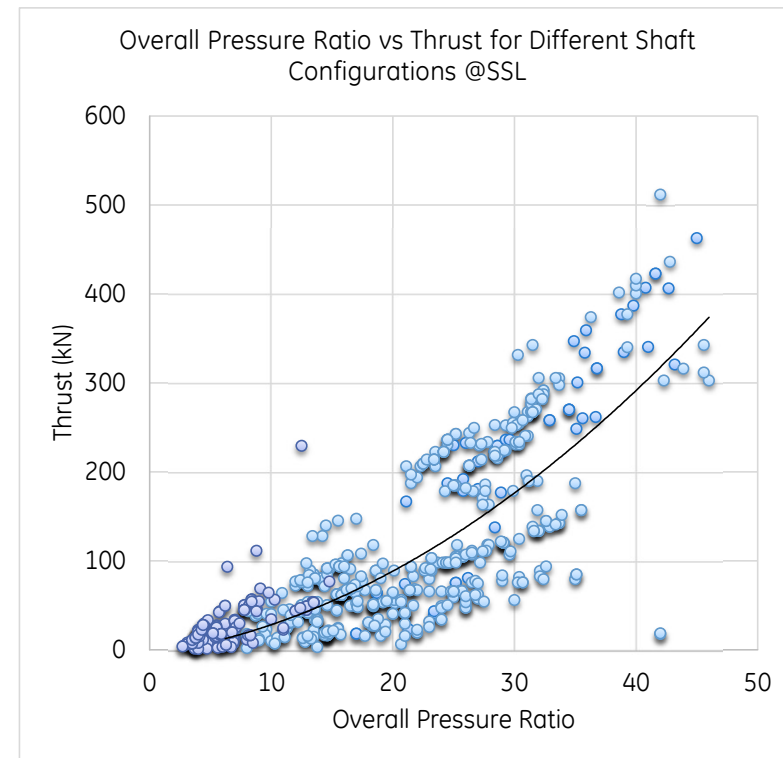
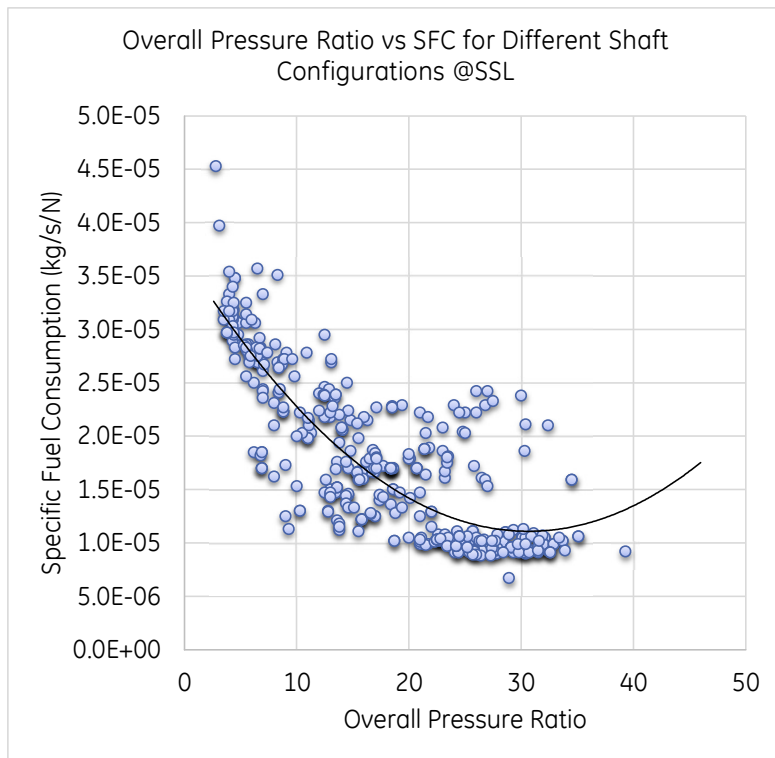
Engine Pressure Ratio

- The engine pressure ratio refers to relationship between the total exit pressure vs the total inlet pressure point in the compressor. Also known as overall pressure ratio. Gives a direct indication of the engine performance and dimensions. This term also applies for any other component where there is a pressure rise.

$$\pi_a = \frac{\text{Total pressure leaving component } a}{\text{Total pressure entering component } a}$$



Engine Performance Parameters



Engine Performance Parameters

Fuel to Air Ratio

- Typically, air flows through the inlet and compressor of the gas turbine engine whereas products of combustion flow through the engine components downstream of a combustion process. Most gas turbine engines use hydrocarbon fuels.
- For a perfect gas, the properties such as enthalpy, can be calculated starting with a polynomial for specific heat, this can be done given the ratio of the mass of fuel burned to the mass of air.

$$f = \frac{\text{mass of fuel}}{\text{mass of air}}$$

- The maximum value of f is 0.0676 for hydrocarbon fuels of composition $(CH_2)_n$

Engine Performance Parameters

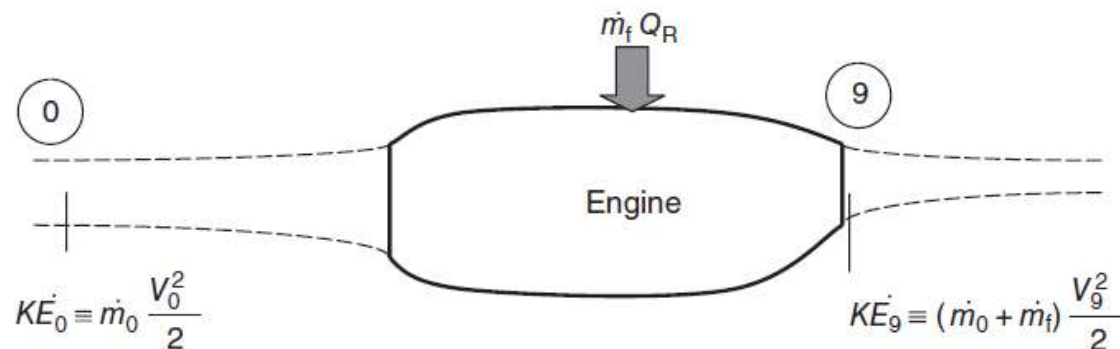
Heating Value

- Available energy per unit of mass of fuel. It is divided in two categories high and low heating value depending on the state of the water present in the combustion products.

Engine Performance Parameters

Thermal Efficiency

- The ability of an engine to convert the thermal energy inherent in the fuel (which is unleashed in a chemical reaction) to a net kinetic energy gain of the working medium is called the engine thermal efficiency

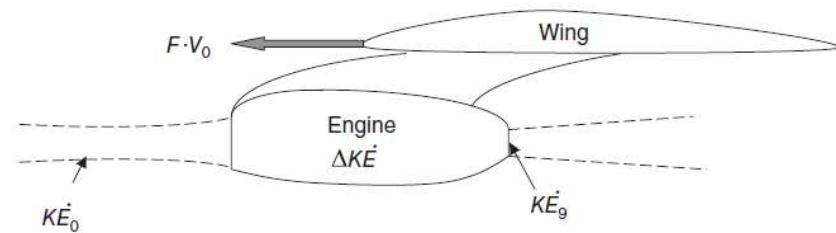


$$\eta_{th} \equiv \frac{\Delta KE}{\dot{m}_f Q_R} = \frac{\dot{m}_9 \frac{V_9^2}{2} - \dot{m}_0 \frac{V_0^2}{2}}{\dot{m}_f Q_R} = \frac{(\dot{m}_0 + \dot{m}_f) V_9^2 - \dot{m}_0 V_0^2}{2 \dot{m}_f Q_R}$$

Engine Performance Parameters

Propulsive Efficiency

- The fraction of the net mechanical output of the engine which is converted into thrust power is called the propulsive efficiency. The net mechanical output of the engine is ΔKE for perfectly expanded nozzle and the thrust power is FV_0 , therefore, the propulsive efficiency is defined as their ratio



$$\eta_p \equiv \frac{F \cdot V_0}{\Delta KE}$$

$$\eta_p \approx \frac{[(\dot{m}_0 + \dot{m}_f)V_9 - \dot{m}_0 V_0]V_0}{(\dot{m}_0 + \dot{m}_f)\frac{V_9^2}{2} - \dot{m}_0 \frac{V_0^2}{2}}$$

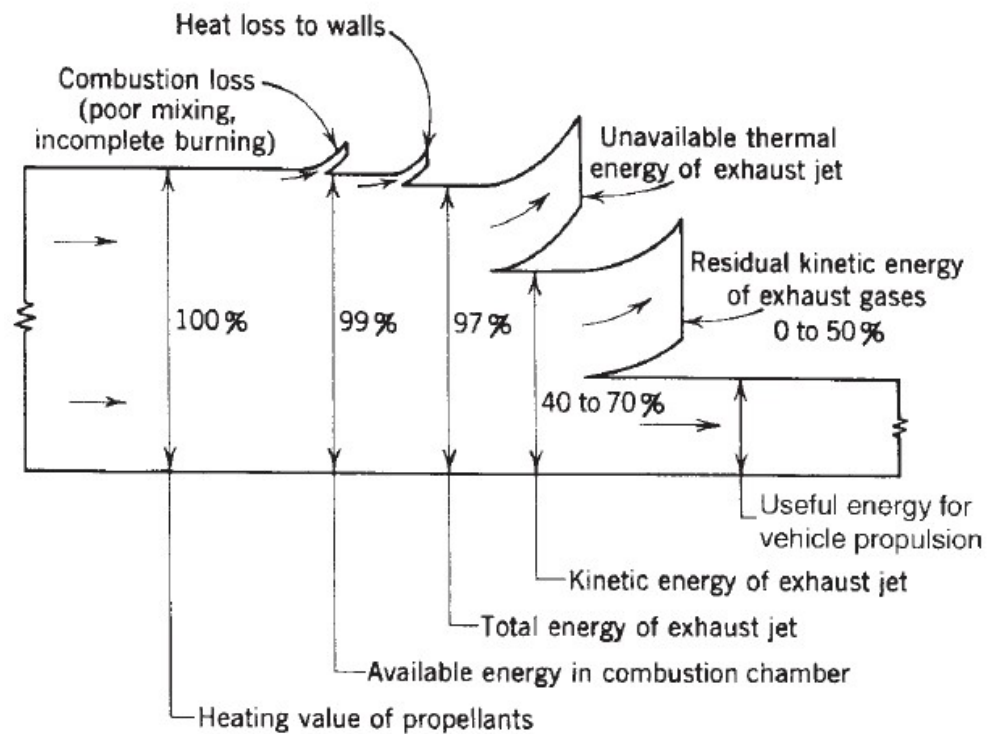
Engine Performance Parameters

Overall Efficiency

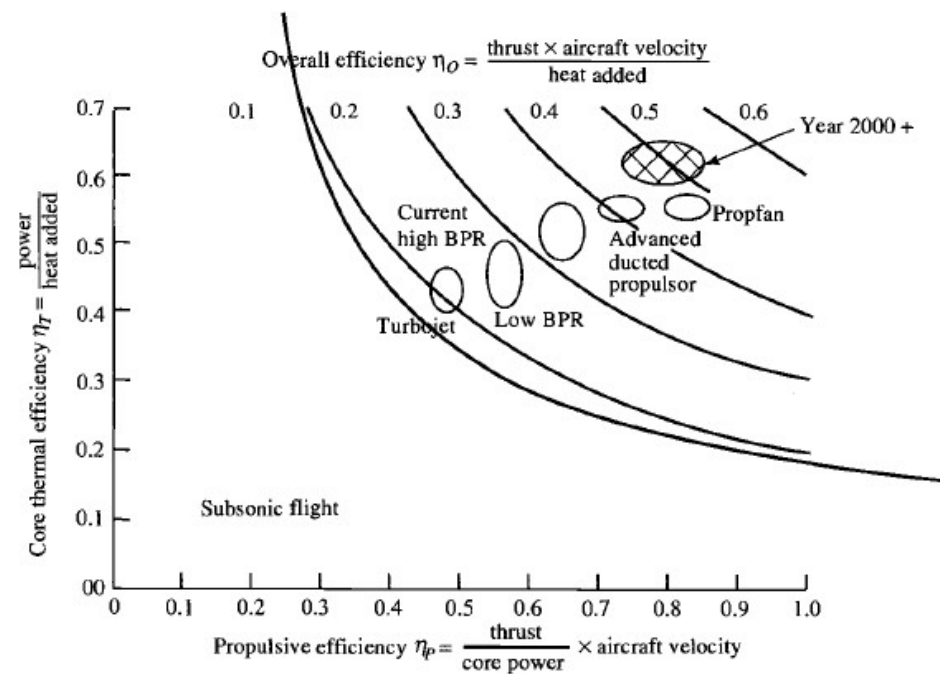
- The overall efficiency of an aircraft engine is therefore the fraction of the fuel thermal power that is converted into the thrust power of the aircraft. Again a useful output is compared with the input investment in this efficiency definition.

$$\eta_0 \equiv \eta_{\text{th}} \cdot \eta_p = \frac{\Delta KE}{\dot{m}_f Q_R} \frac{F \cdot V_0}{\Delta KE} = \frac{F \cdot V_0}{\dot{m}_f Q_R}$$

Engine Performance Parameters



Engine Performance Parameters





Aircraft Design for Propulsion Engineers

Engine Data

- Before a detailed performance estimation can be made it is essential to have representative engine performance charts (or data) available. From the problem definition phase either the engine or the engine type may be known.
- The initial sizing work will have provided an estimate of the engine take-off thrust. To undertake aircraft performance calculations, it is necessary to know what **thrust (and SFC)** the engine will give at thrust settings other than take-off (e.g., at continuous climb, cruise, etc.). It will also be necessary to know the effect of aircraft altitude and speed on the engine parameters.
- For some military aircraft it is also necessary to understand what effect the use of reheat (afterburning) will have on engine performance.

Engine Data

CFM56-5A1

Manufacturer: CFM International : GE + Snecma

Year: 1987

Application: A320-211/-311

Composition: 1 / 3B / 9 / 1 / - / 4

Nb of shafts = 2

$D_{fan} = 1.735 \text{ m} = 68.3 \text{ in}$ $D = 1.829 \text{ m}$

$L = 2.423 \text{ m}$

$W_{eng} = 2266 \text{ kg} = 4996 \text{ lb}$ $= 72 \text{ in}$

$= 95.4 \text{ in}$

Static Sea Level:

Cruise:

$T_{ssl} = 111205 \text{ N}$ $T_{ssl}^{AB} = \text{N}$
 $= 25000 \text{ lbf}$ $= \text{lbf}$

$T_{cr} = 22241 \text{ N}$
 $= 5000 \text{ lbf}$

$SFC_{ssl} = 0.93 \cdot 10^{-5} \text{ (kg/s)/N}$ $SFC_{ssl}^{AB} = \text{(kg/s)/N}$
 $= 0.33 \text{ (lb/h)/lbf}$ $= \text{(lb/h)/lbf}$

$SFC_{cr} = 1.69 \cdot 10^{-5} \text{ (kg/s)/N}$
 $= 0.6 \text{ (lb/h)/lbf}$

$\dot{w}_{ssl} = 386.5 \text{ kg/s} = 852.1 \text{ lb/s}$

$\dot{w}_{cr} = \text{kg/s} = \text{lb/s}$

BPR = 6 OPR = 26.5

OPR_{cr} = 31.3

FPR = 1.55 TET = 1537 °K

$M_{cr} = 0.8$, $h_{cr} = 10668 \text{ m}$

Engine's parameters impact on Aircraft Performance

- Fuel Consumption/Fuel Fraction
- The rate of change of the aircraft weight dW/dt is due to the fuel consumed by the engines. The mass rates of fuel consumed is equal to the product of the installed thrust T and the installed thrust specific fuel consumption.

$$\frac{dW}{dt} = -\dot{w}_f = -\dot{m}_f g_0 = -F_n(TSFC)g_0$$

- Or dimensionless

$$\frac{dW}{W} = -\frac{F_n}{W} (TSFC)g_0 dt$$

Engine's parameters impact on Aircraft Range and Endurance

- Range

- For portions of aircraft flight where distance is important the differential in time dt is related to the differential distance ds by the flight speed, which combined with the fuel consumption equation leads to

$$R = \left(M_o \frac{L}{D} \right) \frac{a_0 / g_c}{TSFC} \ln \frac{W_i}{W_f}$$

- Endurance

- For portions of aircraft flight where distance is important the differential in time dt is related to the differential distance ds by the flight speed, which combined with the fuel consumption equation led to

$$E = \left(\frac{L}{D} \right) \frac{1 / g_c}{TSFC} \ln \frac{W_i}{W_f}$$

Engine Size and Selection

- Thrust to weight ratio
 - Thrust-to-weight ratio is a dimensionless ratio of thrust to weight of a rocket, jet engine, propeller engine, or a vehicle propelled by such an engine that is an indicator of the performance of the engine or vehicle.

- From T/W ratio

$$F_{Installed_{TO}} = T/W \times M_{TO} \times g_0$$

- Maximum take-off mass

$$M_{TO} = \frac{M_{UL}}{1 - (M_E/M_{TO}) - (M_F/M_{TO})}$$

Where:

M_{TO} = maximum take-off mass

M_{UL}^* = mass of useful load (i.e payload, crew and operational items)

M_E^* = empty mass

M_F = fuel mass

*When using the above equation, it is important not to double account for mass components.

Thrust to Weight Ratio - Data

	<i>PAX</i>	<i>Range</i> (nm)	M_{TO} (kg, lb)	<i>T/W</i>	<i>W/S</i> (kg/sq.m, lb/sq.ft)	M_E / M_{TO}
<i>Business jets</i>						
Falcon 2000	19	3000	15 875, 35 000	0.327	323, 66.3	0.563
Gulfstream V	14	6500	40 370, 89 000	0.332	382, 78.3	0.526
Learjet 45	10	2200	8 845, 19 500	0.359	359, 73.6	0.600
Canadair RJER	50	2270	23 133, 48 800	–	478, 98.1	0.591
Beechcft 400A	8	1690	7 303, 16 100	0.360	326, 66.8	0.624
Hawker 100	10	3010	14 061, 31 000	0.340	404, 82.8	0.581
Citation	11	3300	15 650, 34 500	0.371	–	0.586
<i>Commercial jets</i>						
Fokker 100	107	1680	44 450, 98 000	0.308	475, 97.3	0.556
Romero 1-11	109	1480	47 400, 104 500	0.289	494, 101.2	0.500
RJ100	112	2090	44 000, 97 000	0.290	572, 117.2	0.573
B717-200	106	1460	49 895, 110 000	0.291	536, 110.0	0.614
A318-100	107	2350	64 500, 142 200	0.330	526, 107.8	0.627*

Engine Size and Selection - 787

GE9x-1B70

Manufacturer: GE : General Electric

Application: B787

Composition: 1 / 4B / 10 / 2 / - / 7

$D_{fan} = 2.82 \text{ m} = 111 \text{ in}$ $D = \text{m}$

$W_{eng} = \text{kg} = \text{lb}$ $= \text{in}$

Static Sea Level:

$T_{ssl} = 311000 \text{ N}$ $T_{ssl}^{AB} = \text{N}$

$= 69916 \text{ lbf}$ $= \text{lbf}$

$\text{SFC}_{ssl} = (\text{kg/s})/\text{N}$ $\text{SFC}_{ssl}^{AB} = (\text{kg/s})/\text{N}$

$= (\text{lb/h})/\text{lbf}$ $= (\text{lb/h})/\text{lbf}$

$\dot{w}_{ssl} = \text{kg/s} = \text{lb/s}$

BPR = 9.6 OPR = 43

FPR = TET = °K

Year: 2008

Nb of shafts =

$L = \text{m}$

$= \text{in}$

Cruise:

$T_{cr} = \text{N}$

$= \text{lbf}$

$\text{SFC}_{cr} = (\text{kg/s})/\text{N}$

$= (\text{lb/h})/\text{lbf}$

$\dot{w}_{cr} = \text{kg/s} = \text{lb/s}$

OPR_{cr} =

$M_{cr} = , h_{cr} = \text{m}$

Trent 1000

Manufacturer: Rolls-Royce

Application: B787-3/-8/-9, A350-800/-900 (expected)

Composition: 1 / 8 / 6 / 1 / 1 / 6

$D_{fan} = 2.845 \text{ m} = 112 \text{ in}$ $D = \text{m}$

$W_{eng} = 6437 \text{ kg} = 14191 \text{ lb}$ $= \text{in}$

Static Sea Level:

$T_{ssl} = 235755 \text{ N}$ $T_{ssl}^{AB} = \text{N}$

$= 53000 \text{ lbf}$ $= \text{lbf}$

$\text{SFC}_{ssl} = (\text{kg/s})/\text{N}$ $\text{SFC}_{ssl}^{AB} = (\text{kg/s})/\text{N}$

$= (\text{lb/h})/\text{lbf}$ $= (\text{lb/h})/\text{lbf}$

$\dot{w}_{ssl} = 1247.4 \text{ kg/s} = 2750 \text{ lb/s}$

BPR = 10.8 OPR =

FPR = TET = °K

Year: 2008

Nb of shafts = 3

$L = 3.861 \text{ m}$

$= 152 \text{ in}$

Cruise:

$T_{cr} = \text{N}$

$= \text{lbf}$

$\text{SFC}_{cr} = (\text{kg/s})/\text{N}$

$= (\text{lb/h})/\text{lbf}$

$\dot{w}_{cr} = \text{kg/s} = \text{lb/s}$

OPR_{cr} =

$M_{cr} = , h_{cr} = \text{m}$